

Where the network binds, dispatching becomes road scheduling

Deadlock-free task assignment and passage sequencing: pushing up the lower edge of the capacity interval (research plan)

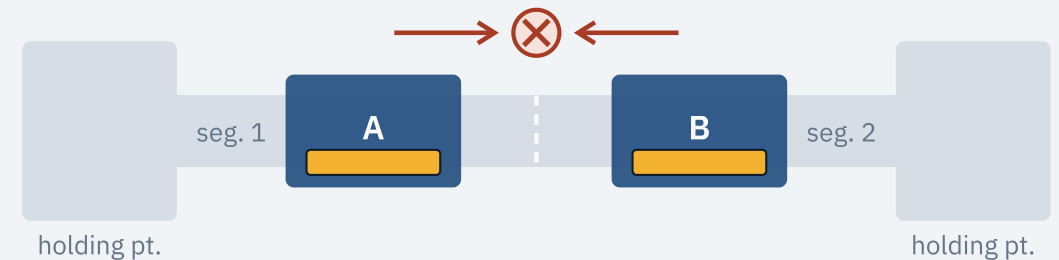
Deadlock-free task assignment and traffic orchestration for agents whose footprint grows when loaded (working title)

Chunrui Na

Harbin Engineering University · Nanyang Technological University

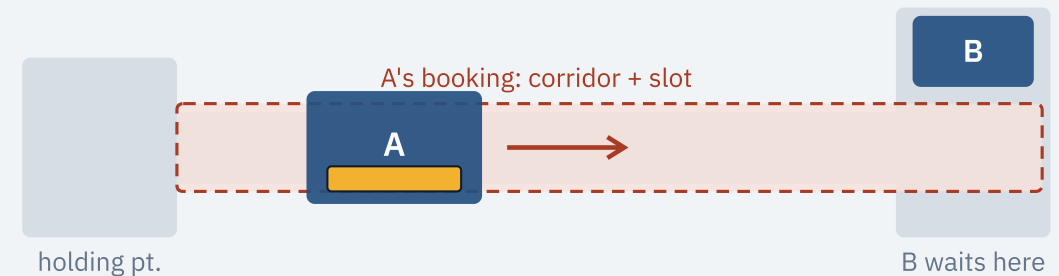
Grounded in chapter 3's reviewed results (thesis-core a882742: T18–T23) and research manifesto v1

Naive segment request: head-on in one lane



A holds 1, waits for 2; B holds 2, waits for 1

T5: reserve up to the next safe holding point



The slot is booked before release: no hold-and-wait cycle

This study in one minute

1

The question

Capacity is an interval, [best deadlock-free rule, T1']. Okpo's main scenario gives [627, 1,720] a day: the lower edge is whole-route reservation, only 36% of T1'.

2

Starting point (T22)

Safe release, segment_safe, is built: 860,291 decisions match a naive reduction, and it never deadlocks; but about 2/3 of its waits start from a criterion refusal, and from K = 20 on it stays below reservation.

3

The approach

T27 is running the first rule comparison: Banker plus a step-wise feasibility search, and claims cut at the next safe holding point (T5). Then assignment and passage order are searched together inside that safety shell.

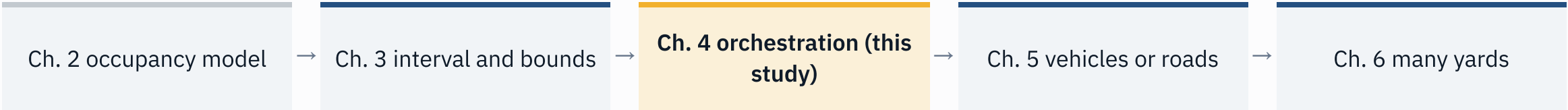
4

Outputs

A lower edge pushed towards field fleets, which under a 24-h day sit between free flow and reservation; an answer to H5, what safety costs; a proof of T5; a chapter 4 journal paper.

Where the network binds, how much of the bound is achieved is decided by traffic orchestration. (C3)

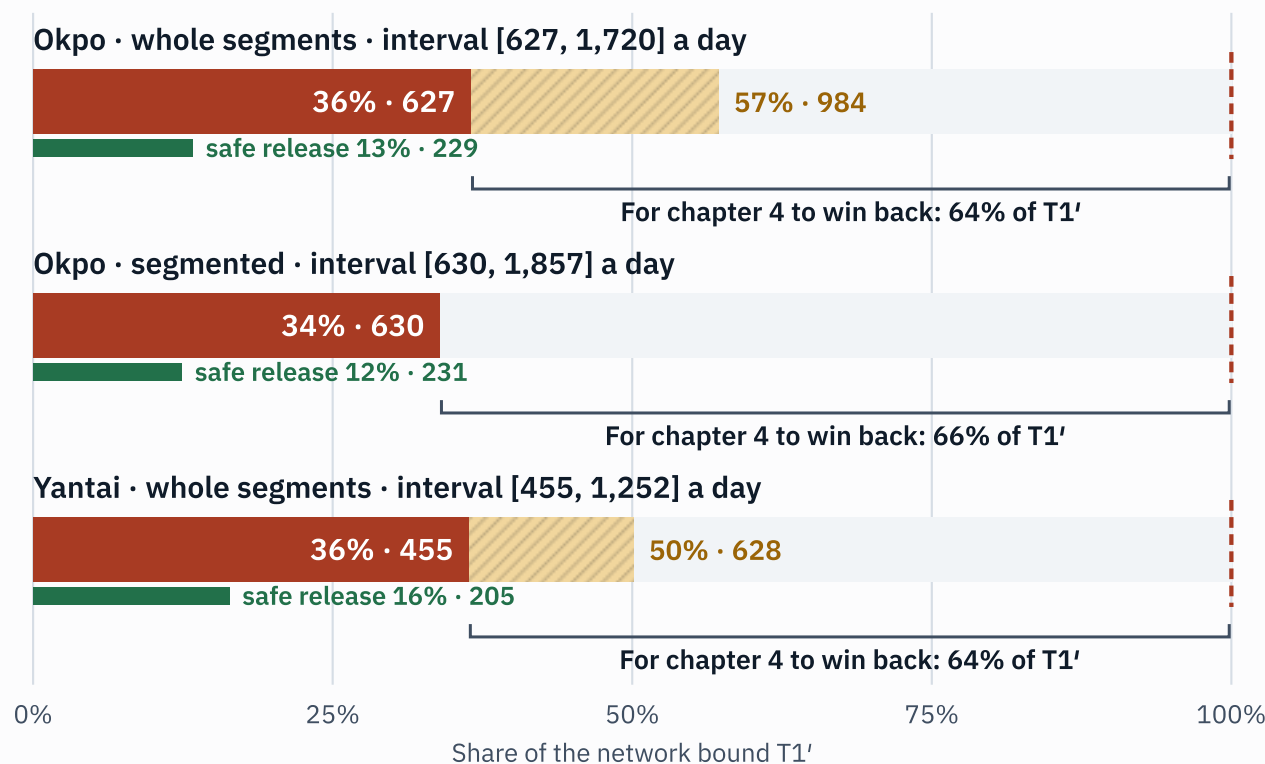
Its place in the thesis: chapter 3 gives the capacity interval, chapter 4 pushes its lower edge up



ITEM	MANIFESTO V1 WORDING	HOW CHAPTER 4 ANSWERS
C1 interval	Achievable capacity lies in the interval [best deadlock-free rule, network bound]	Today's lower edge is whole-route reservation (T22 also tested safe release, which is more conservative); the rules chapter 4 delivers take over, starting with T27
C3 dispatching becomes road scheduling	Where the network binds, how much of the bound is achieved is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks	All four levers become decision variables; deadlocks are counted only under no-teleport rules
C4 boundary	When the plateau lies far below the bound, change the orchestration first. This is the boundary between C3 and C4	Okpo's main scenario: plateau / $T1' = 0.36$ (segmented 0.34), so orchestration comes first; erection mix 0.87 with whole segments, 0.53 segmented (T22, page 5)
H5 safe orchestration	Naive dispatching deadlocks or nearly deadlocks at peaks; a provably safe policy loses only a little throughput	First reading on page 10 ($K = 30$: reservation 61%, safe release 46%); T27 re-checks it
T5 theorem	Safe orchestration: reserving to the next safe holding point never deadlocks, with a loss bound relative to the upper bound	Manifesto status “to do (chapter 4)”; T27 is testing it (running); proof route on page 9

Sources: research manifesto v1 (thesis-core tag manifesto-v1, docs/研究宣言.md §§2–4), Chinese wording quoted verbatim and translated here; readings from the T22 report in thesis-core a882742, §§4, 6.3; chapter links from the thesis direction document (2nd ed.), §4.

The capacity interval [best deadlock-free rule, T1']: on both yards the lower edge is whole-route reservation, about 36% of T1'



36% / 36%

Whole-route reservation plateau ÷ T1' (Okpo / Yantai, whole segments): the best deadlock-free rule, i.e. the lower edge

13% / 16%

Safe release, segment_safe ÷ T1': also deadlock-free with no teleporting, but more conservative (page 8)

57% / 50%

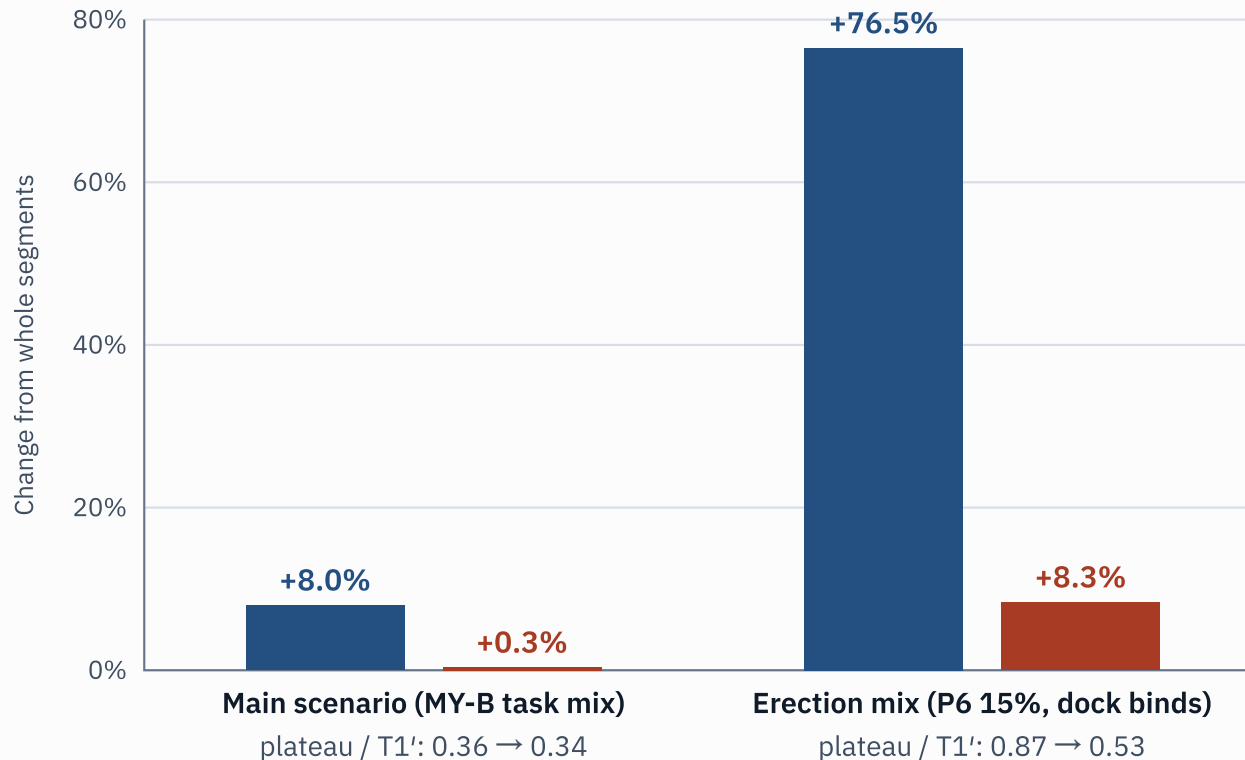
Segment request (reference rule, teleport clearing) ÷ T1': it relies on teleporting, so it cannot be the lower edge

■ Whole-route reservation: best deadlock-free rule = lower edge ■ Safe release, segment_safe

■ Segment request (reference rule, teleport clearing) [] T1' network bound

Sources: thesis-core a882742, T22 report §4 (t22_interval.csv). Whole segments: Okpo results/a0ab16fceb94 and 49d2383872e7 (MY-B task mix), Yantai results/b82aa19b22af; safe release and segmented roads: main_safe, Okpo 656e2da75368, Yantai 930e637e0658. Conditions: saturated, 16-h day, 10 seeds; plateau = mean throughput over K = 100–150, T1' = mean bound over 10 seeds. Maps: Okpo rev 1825 (the T18 batches use rev 1824, an identical network), Yantai r396 (T20 batches r395, identical network); from T27 on Okpo moves to rev 1833 from 20_数据 (floating docks, branch roads), while every reading in this briefing is on rev 1825 / r396.

The rule, not the road, limits the main scenario: segmenting the dock roads lifts T1' by 8%, the lower edge by 0.3%



The C3–C4 boundary: two ready examples

Main scenario: the bound rises, the lower edge barely moves, so orchestration comes first. Erection mix (the dock binds): whole-segment reservation already reaches 87% of the bound, so road changes pay; segmented, the bound gains 76.5%, reservation 8.3%, and the ratio falls to 53%: the bottleneck moves to the rule.

Small instances (checked by hand)

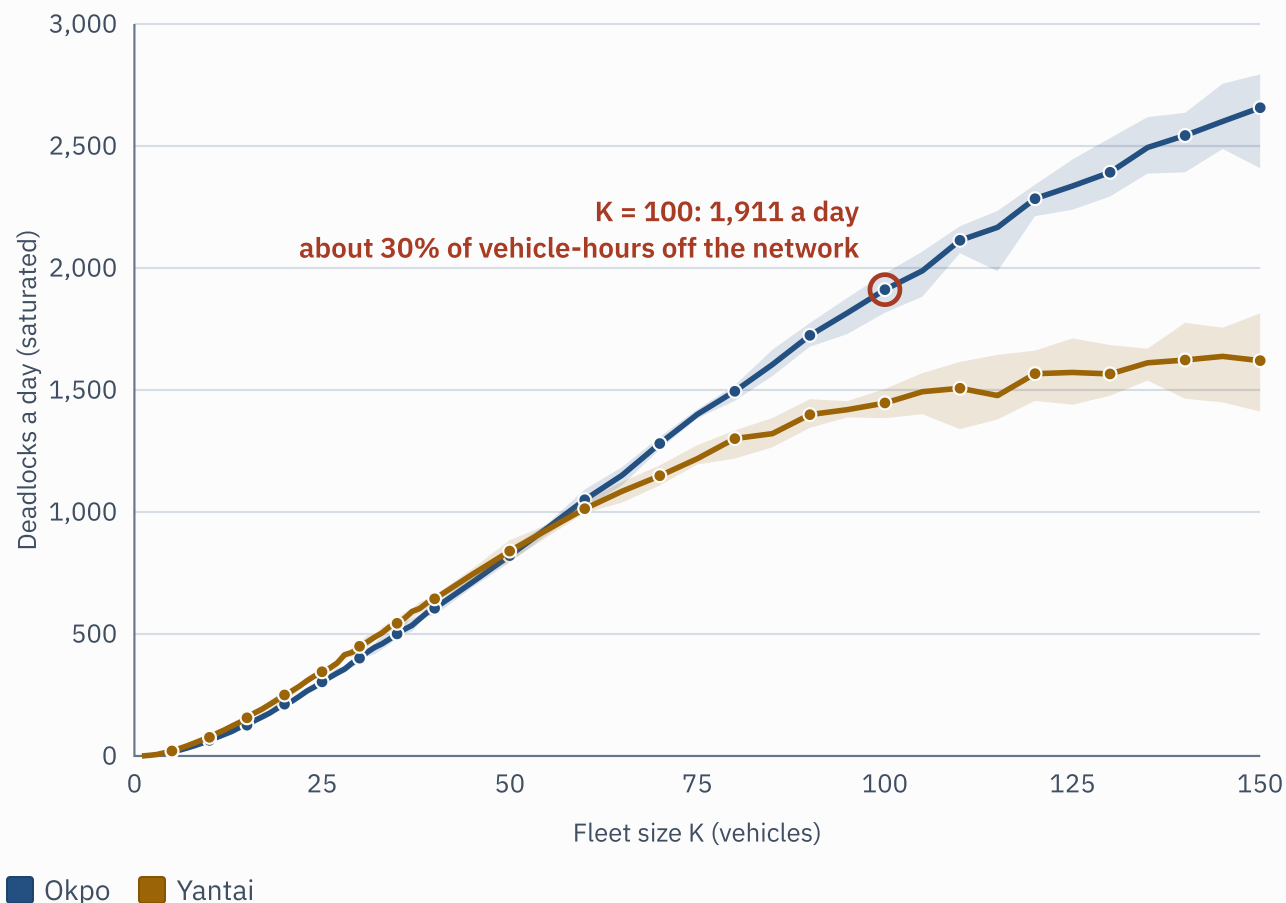
A 12 m single-lane corridor: reservation 19 a day, a hand-built feasible schedule 31.6, T1' 37.9. A crossroads: reservation 70, segment request (reference rule, teleport clearing) 105, T1' 187.

The cause is a rigid chain: reservation books empty run, loading, loaded run and unloading as one chain, so booked road idles during handling. Widening and passing bays get 10-seed numbers in T25 (3-seed probe: 12 m lifts T1' 5%, reservation 0.2–0.9%).

■ T1' network bound ■ Whole-route reservation plateau (lower edge)

Sources: thesis-core a882742, T22 report §§3.5, 4 (t22_interval.csv). Main scenario: whole segments as on page 4, segmented main_safe 656e2da75368; erection mix: whole T18 f7ac88b60111 and dfb2758822ad, segmented erect_safe a1f485302418. Segmenting rule: roads longer than 200 m with roadside stops get block boundaries 25 m either side of each stop (a scenario setting; T25 scans block length); on both yards the rule picks exactly the dock roads. Plateau = mean over K = 100–150; T1' is the mean of 10 seeds. Small instances and the rigid chain: T18 report §2.3; the reviewer's probe: the author's guidance of 9 Oct 2026, appendix A.

Naive segment request deadlocks; it survives only by “teleport” clearing, so it is a reference rule



26 / 83

Okpo deadlocks a day on a regular / peak day (97 / 194 tasks, the 4-vehicle subset of Yim 2008, for comparison only)

1,911 / 2,657

Deadlocks a day at saturation, K = 100 / 150

≈ 30%

Vehicle-hours teleported off the network at K = 100, at least (478 / 1,600)

T22 now reads physical rules: safe release never deadlocks; with no clearing the whole network gridlocks, faster as K grows (Okpo: about 4.3 h at K = 5, under 1 h from K = 40). Under the reference rule about 40–50% of loaded P6 moves are teleported at least once on the way to the dock, so it serves as comparison only. Support from ports: a terminal network deadlocks from 30 vehicles without keep-clear hatching at junctions (Zhou et al. 2017).

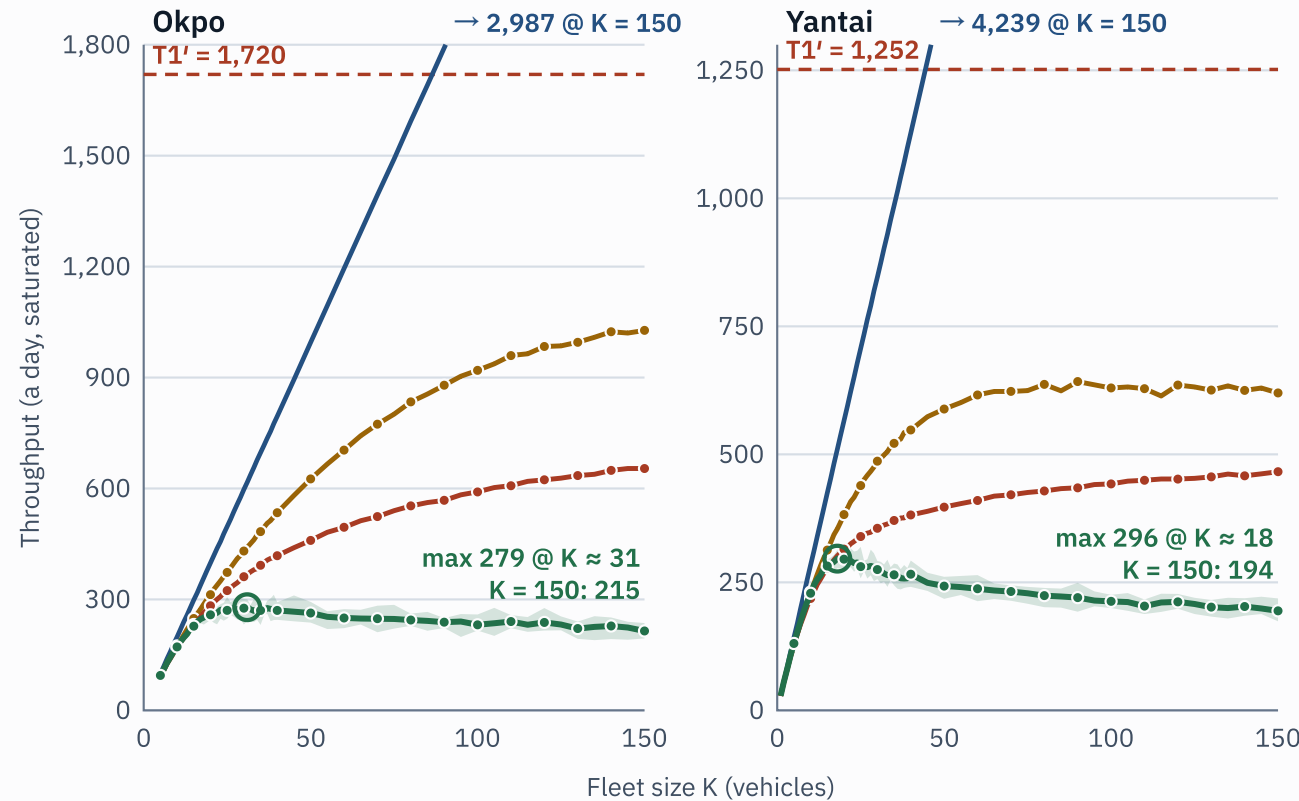
Sources: thesis-core a882742, Okpo T18 a0ab16fceb94 and 49d2383872e7, Yantai T20 b82aa19b22af (conditions as on page 4). Segment request (reference rule, teleport clearing): at each deadlock one vehicle releases all its resources, waits 900 s on open ground and requests again from where it was (no published value for 900 s, T17). Deadlocks are means of 10 seeds, shading min–max; regular and peak are medians over the K grid. Vehicle-hours off the network = deadlocks × 900 s ÷ (K × 16 h), a lower bound. No clearing and teleports counted per move: T22 report §§3.4, 5.

Rules in use and in progress: reservation is today's lower edge, safe release is built, T27 is trying two wider ones

RULE	HOW IT RELEASES	DEADLOCK	LIMITATION	USE IN CHAPTER 4	STATUS
Whole-route reservation	Books the whole route before a loaded departure	none	Rigid chain; plateau about 36% of T1'	Today's best deadlock-free rule: the lower edge	done, T18
Segment request (reference rule, teleport clearing)	One segment at a time; at a deadlock one vehicle leaves for 900 s	yes	Outside the T1' proof; 40–50% of loaded P6 moves teleported	Reference only	done, T17
Segment request, no clearing	As above; deadlocked vehicles keep the road	yes	The whole network gridlocks, faster as K grows	Measures naive deadlocks (H5, first criterion)	done, T22
Safe release, segment_safe	Grants a request only if the system stays in a safe state (Banker-type)	none	Claims whole remaining tasks: 2/3 of waits start from refusals; below reservation for $K \geq 20$	Exact, 1.1–2.4× run time; chapter 4's start	done, T22
Segmented blocking + direction locks	Long roads cut at their stops; one lane: follow one way, the other waits	no head-on	U-turns at stops bypass the lock (under 1% of deadlocks); fixed in T27 step 0	Block-granularity lever; Okpo's main scenario segmented	done, T22
T27: step-wise push; T5	Grant if Banker or a step-wise push succeeds; T5: depart only with all road to the next safe holding point	none (T5 to prove)	Holding-point placement and capacity set the loss	Wider, just as reliable; reports the refusal share	T27 running

Sources: the T17, T18 and T22 reports in thesis-core a882742 (T22 §§2.1–2.3, 3.4, 5); the T27 task note (00_总控/runs/2026-10-10_T27_prompt.md; running, no results yet); the author's guidance of 9 Oct 2026, §2.2, and imagery check (neighbouring sites stay reachable while a vehicle works at the dock edge of road 161, so from T22 on Okpo's main scenario is segmented at the stops, with whole segments as comparison); T5 wording per manifesto v1.

Chapter 4's starting point: safe release is exact and never deadlocks, yet from $K = 20$ on it stays below whole-route reservation



■ Safe release, segment_safe
 ■ Whole-route reservation
■ Segment request (reference rule, teleport clearing)
 ■ Free flow
 - - - $T1'$ network bound

860,291

decisions identical to a naive reduction without caching: 0 mismatches, 0 deadlocks

$\approx 2/3$

of safe-release waits begin with a criterion refusal while capacity is free (main scenario, all K : Okpo 61%–75%, Yantai 60%–70%)

279 → 215

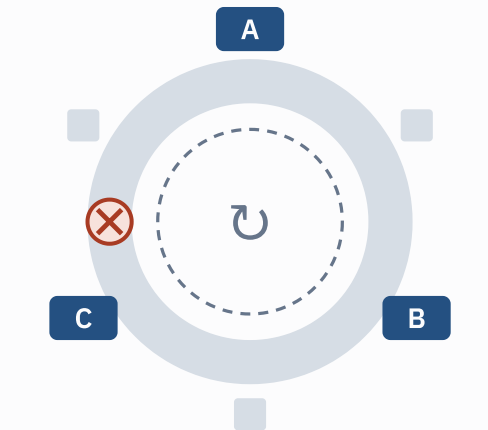
Okpo: max 279 @ $K \approx 31$, 215 at $K = 150$; Yantai 296 @ 18 → 194. The rise and fall belong to this criterion; T27 re-checks them

Where it is conservative: the check assumes the others stand still and each vehicle finishes its whole remaining task in turn, so vehicles that would never meet can be judged unsafe (ring6: three vehicles circling one way; the reference rule never deadlocks, safe release refuses once). Refused vehicles wait on the road. Run time: 1.1–2.4× the reference rule.

Sources: thesis-core a882742, T22 report §§2.1, 4, 5 (t22_summary.json, t22_naive.json). Safe release: main_safe, Okpo results/656e2da75368, Yantai 930e637e0658 (scenario main, whole segments); other rules: Okpo T18 a0ab16fceb94 and 49d2383872e7, Yantai T20 b82aa19b22af. Saturated, 16 h, 10 seeds; the green shading is safe release's min–max; conditions as on page 4. Free flow ignores occupancy and is not drawn beyond the frame.

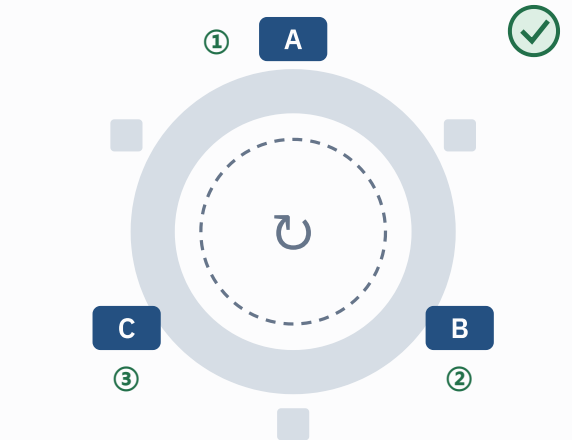
T27 (running): chapter 4's first rule comparison tries two releases wider than Banker and just as reliable

(a) Banker check (T22 safe release)



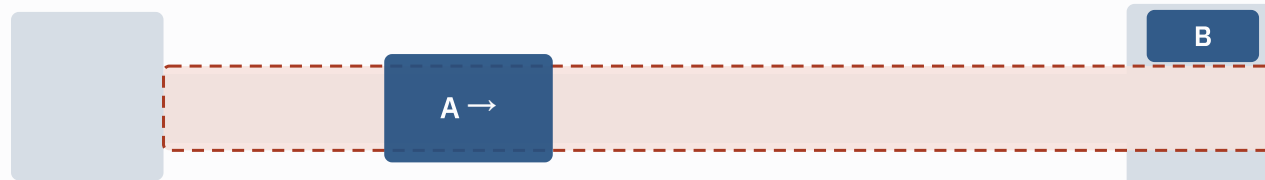
Each route crosses the next car's entrance
→ judged unsafe: C refused (ring6)

(b) Step-wise feasibility push



Push one decision point at a time on a copy
→ all three finish: grant; the pushes are the plan

(c) T5: claims cut at the next safe holding point

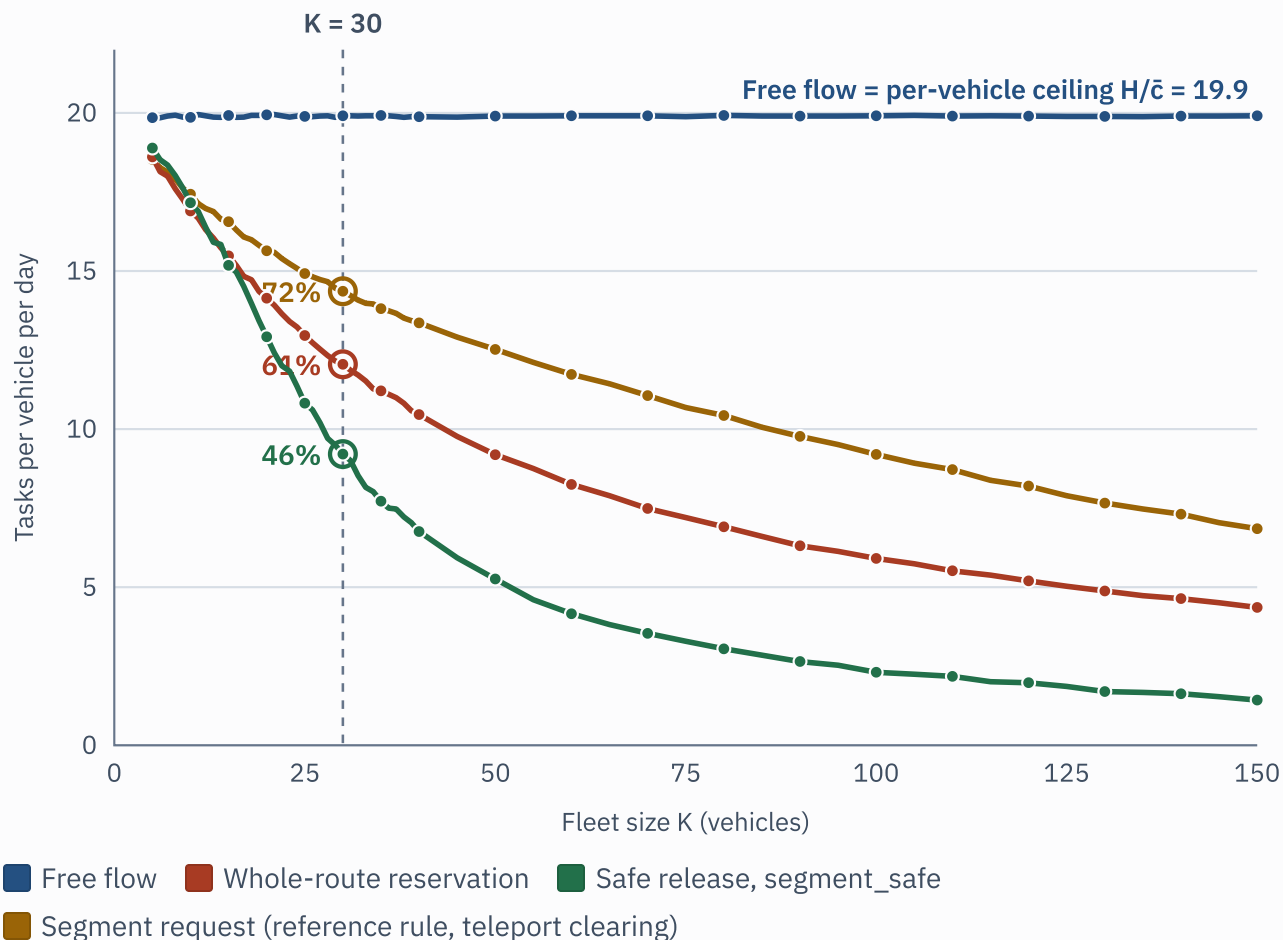


A books the corridor and a slot at the far end; B waits there: no hold-and-wait cycle

What T27 does (dispatched 10 Oct 2026; running, no results yet)

- **Two new rules:** grant if Banker holds or a step-wise push carries every vehicle to the end of its task (segment_push: as reliable, never more conservative); or depart only with everything up to the next safe holding point (segment_haven, i.e. T5).
- **Readings:** each rule's share of waits caused by criterion refusals; re-checks of the lower edge, H2's decline and H5's cost of safety (1 – best safe plateau / T1').
- **Step 0:** maps from 20_数据 (Okpo rev 1833); U-turns at stops under the direction lock; reservation waits split into road and crane; resource names disambiguated.
- **Outputs and proof route:** the chapter 4 rule-comparison figure, the lower edge re-checked, and T5's sufficient condition with a draft proof (the author writes the formal one). T5 meets T1's (A1)–(A3); max-plus gives its loss for a given order.

A first reading of H5: at $K = 30$, per-vehicle efficiency is 61% of the free-flow ceiling under reservation and 46% under safe release



61% / 46%

Per vehicle per day ÷ free-flow ceiling at $K = 30$: reservation / safe release; the reference rule (teleport clearing) reaches 72%

64% / 64%

The best safe rule's loss against $T1'$ (reservation; Okpo / Yantai), the manifesto criterion; the teleporting reference rule still loses 43% / 50%

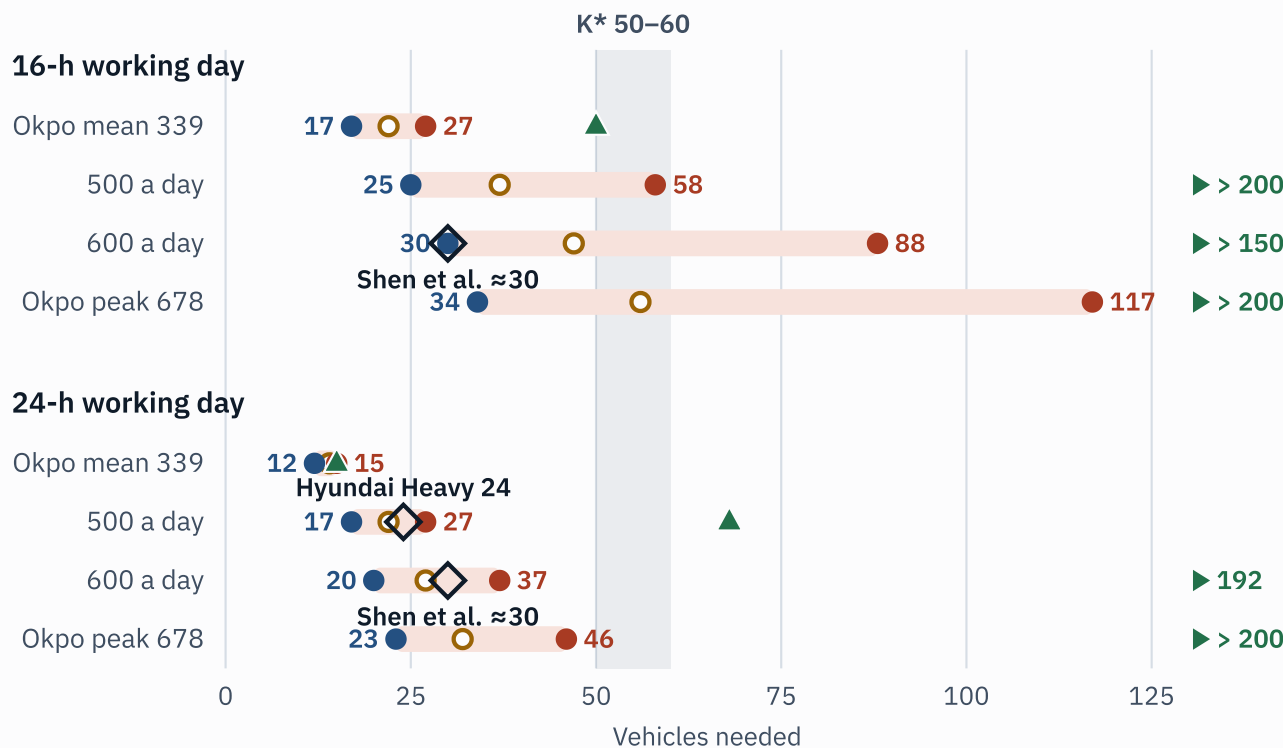
H5 safe orchestration (manifesto v1)

Criteria: deadlocks are counted under no-teleport rules; the cost of safety is the best safe rule's loss relative to the network bound. **If it holds:** safe orchestration is cheap. **If not:** a formula for the cost of safety and its conditions.

Efficiency is normalised by cycle time and meets the field once the anchor yards' haul distances are known; the loss against $T1'$ is the manifesto criterion. T27 re-checks both.

Sources: thesis-core a882742, T22 report §6.3 (t22_anchor.csv: Okpo, 16 h, $K = 30$) and the main-scenario saturated curves (batches as on page 8). Free-flow ceiling $H/\bar{c} = 960 \text{ min} \div \text{free-flow cycle of } 48.3 \text{ min}$ (empty 9.1, loading 10, loaded 19.1, unloading 10). Loss against $T1' = 1 - \text{plateau} / T1'$ (page 4). H5 per manifesto v1; the reading "cost of safety = $1 - \text{best safe rule's plateau} / T1'$ " follows the T27 task note.

Whole-yard vehicles needed: what a better rule could save lies between reservation and free flow



88 → 30

Okpo at 600 a day, 16 h: whole-route reservation 88 vehicles, reference rule 47, free flow 30 (24 h: 37 / 27 / 20)

27 / 15

Okpo's own mean of 339 a day: 16 h reservation 27, free flow 17; 24 h 15 / 12, all below $K^* = 50-60$

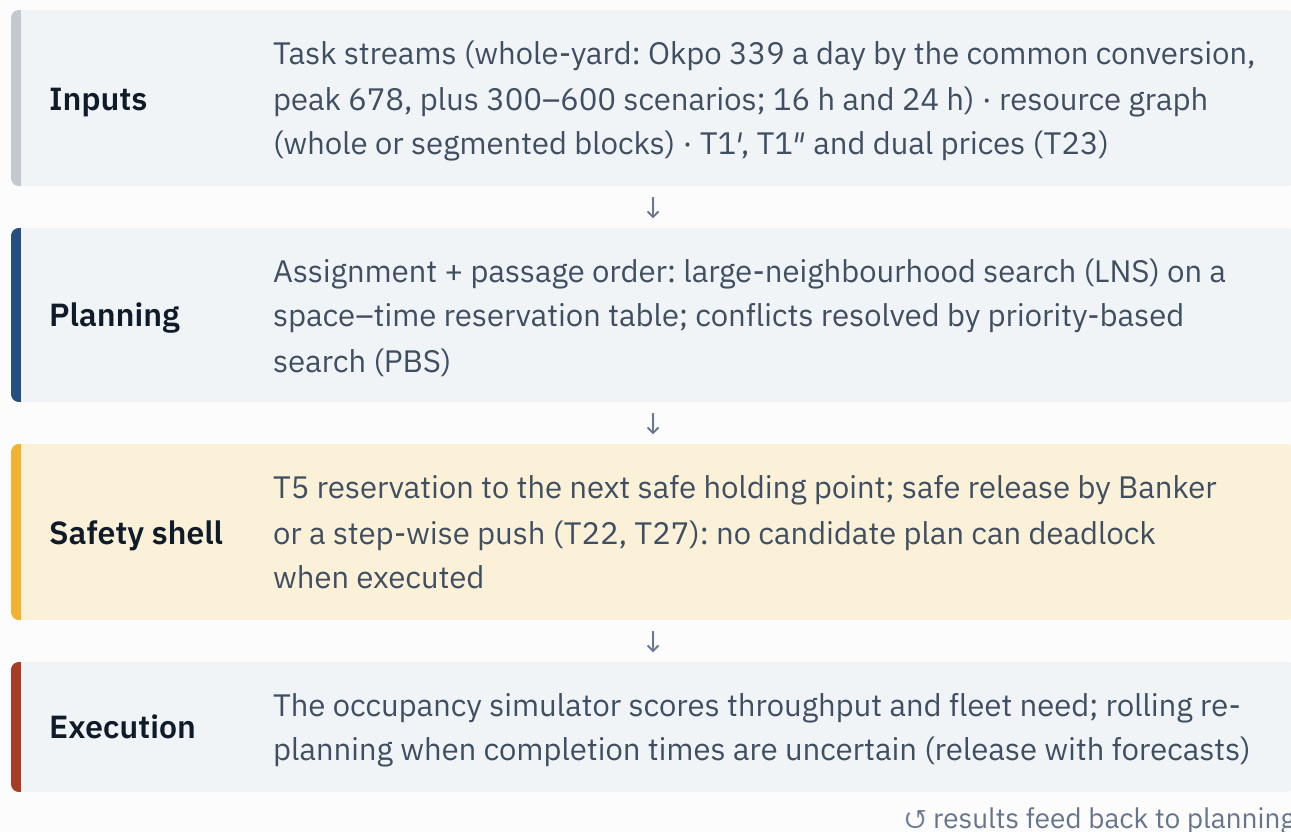
Where field fleets sit

Hyundai Heavy (about 500 moves a day, 24 h) runs 24 vehicles, between the model's 24-h free flow (17) and reservation (27); Shen et al. (about 600 a day, shift not stated) report about 30: between 20 and 37 at 24 h, while at 16 h only free flow (30) matches. At 24 h, real fleets sit between the two.

The main caliber (16 or 24 h, mean or peak) is the author's decision after T27; both are shown. At 16 h safe release misses 500 a day and above up to $K = 150$ or 200 (arrows).

Sources: thesis-core a882742, T22 report §§6.1–6.2 (t22_demand.csv; Okpo whole segments: demand c6b4a59a7a82, demand_t24 0fb7e9aefc36, demand_peak a79d535a75b3). Vehicles needed = smallest integer K whose mean throughput over 10 seeds reaches 99% of the daily volume (4 days, day 1 warm-up); 339 is Okpo's volume by the common conversion (T24), $678 = 339 \times 2$ the peak; $K^* = 50-60$ (T18). Field fleets: library notes/occupancy_anchors_20261009.md §1.8 (commit e483924; 박창규 and 서준용 2006; Shen et al. 2018), taking a real fleet as equal to the vehicles needed (no maintenance or spare margin).

Methods: a safety shell around an integrated search over task assignment and passage order



Ground truth: CP-SAT exact model

One interval variable per move and resource, one no-overlap per segment and junction (cumulative where width is shared); optima on small instances.

Baselines

Whole-route reservation (today's best deadlock-free rule), safe release with T27's new rules, and “schedule in free flow, then repair conflicts”: they price integration and the shell separately.

Starting point: MY-B code

ALNS and CP-SAT from the fleet-sizing paper; its CP-SAT already proves minimum fleets on 12–13-block batches.

[Fleet-sizing \(MY-B\) briefing →](#)

Optional: LLM-evolved rules

EoH, ReEvo, FunSearch (abstract level; no shipyard use found), compared with hand-written rules and optimisation.

Every component has a mature source; what is new is an agent that widens when loaded, long lifts and a capacity bound

COMPONENT	TRANSFER SOURCE AND THEORY	WHAT IS NEW IN ROAD-OCCUPYING SHIPYARD TRANSPORT
Exact model	Job shop with blocking (Mascis & Pacciarelli 2002); OR-Tools CP-SAT	Width-shared resources; asymmetric loaded and empty occupancy; long holds at loading points
Integrated search	MAPF-LNS (Li et al. 2021); ALNS (Ropke & Pisinger 2006); MY-B code	Assignment and passage order searched together; empty runs need road too; fleet size is part of the decision
Conflict resolution	Priority-based search, PBS (Ma et al. 2019); rolling-horizon prioritised planning (Zheng et al. 2026)	Agents whose footprint multiplies when loaded; priorities tied to the reservation mode (whole route, to a holding point)
Safety shell	Banker's algorithm (Habermann 1969); resource allocation systems (Reveliotis 2000, 2005); AGV reservation with consistent priorities (Kim et al. 2006)	Holding points come from the network and vehicle class; the cost of safety is measured against the network bound T1'
Passing and blocking	Kiel Canal traffic with passing at sidings (Lübbecke et al. 2019, abstract level); railway block sections (UIC 406)	Block granularity as an orchestration variable, linked to T25's block-length scan
Optional: rule evolution	EoH, ReEvo, FunSearch (abstract level)	Readable release rules, tested for transfer across yards (links to chapter 6)

Closest in the same field: Tao et al. 2017 (full text) simulate one yard with road capacity counted in vehicles to compare vehicle types, with no blocking-time formalism, rule comparison, deadlocks or bound. Closest in AGVs: Li et al. 2025 (preprint), deadlock-free attachable AGVs via Petri nets and ALNS, which we extend to loads that widen the footprint and to losses measured against the network bound.

Sources: direction document (2nd ed.), §6, §8.5, appendix; guidance of 9 Oct 2026, appendix B; library 03-055 (Tao 2017, T24b), 04-023 (Li 2025) and 04-061 (Zheng 2026), full text. DOIs checked against Crossref. Abstract level: Mascis 2002, Reveliotis 2000 and 2005, Kim 2006, Lübbecke 2019, EoH, ReEvo, FunSearch; metadata only, to be read: Habermann 1969, Ma 2019, Li 2021.

Experiments: each one feeds a figure or table of chapter 4

#	EXPERIMENT	OUTPUT (CHAPTER 4)	NEEDS	STATUS
E1	The deadlock-free lower edge: safe release and reservation × whole / segmented blocks, both yards' saturated curves; gridlock with the no-clearing switch	Fig. 4-1 starting curves and interval; Table 4-1 deadlocks under no-teleport rules (H5, criterion 1)	T22	done
E2	The first rule comparison: step-wise push and T5 (claims cut at the next safe holding point) against Banker and reservation; refusal share; re-check of lower edge, H2 decline and H5	Fig. 4-2 rule comparison; Table 4-2 refusal share and run time	T22; T23	T27 running
E3	Small-instance ground truth and loss split: CP-SAT optima for a corridor, a crossroads and the three-resource dock network; max-plus throughput of T5 for a given order	Table 4-3 gaps between optimum, rules and heuristics; Fig. 4-3 rule loss against bound slack	T23; T25	planned
E4	Integrated search (LNS + PBS + shell) against the three baselines; Okpo 339 / 678 and 300–600 a day, 16 h and 24 h	Fig. 4-4 daily volume against fleet need; Table 4-4 cost of safety (H5, criterion 2)	E2	planned
E5	Holding points and block granularity: spacing, holding capacity S, block length	Fig. 4-5 cost of safety against holding-point spacing and block length	T25; T26	queued after T27
E6	Rolling online: release with forecasts under completion-time forecast errors	Fig. 4-6 throughput against forecast error	E4	planned
E7	External check: vehicles needed and per-vehicle efficiency normalised by cycle time against field anchors (Yim 2008, Heo 2013, Shen 2018, 박창규 2006)	Table 4-5 external check (same basis as chapter 3)	T22; T24	first reading

The next step is a constructive pilot: once T27 names the best safe rule, get CP-SAT and LNS running on E3's small instances, tune parameters and produce the first gap table. Small gaps: scale to both yards. Large gaps: revise the neighbourhoods and the shell, then continue.

Sources: the author's guidance of 9 Oct 2026, §§2.1–2.5, 4.1, 4.3; the T18 (§§2.3, 7), T22 and T23 reports in thesis-core a882742; task board tasks.md, T22–T27 (10 Oct 2026). Figure and table numbers are planned numbers.

What holds, what is running, what is queued, and risks with mitigations

Holds (T22, T23, T24, reviewed)

Both ends of the interval, reservation at the bottom; safe release exact, deadlock-free, conservative; gridlock without clearing; segmenting lifts only the main upper edge; route-free T1" (draft proof); whole-yard fleet need, field anchors

Running (dispatched 10 Oct 2026)

T27: step-wise push and T5, refusal shares, re-checks of lower edge, H2 and H5. Then the author decides three H1 items: part 2's numbers, part 3's two-layer wording, the main caliber (16 / 24 h, mean / peak)

Queued (after T27)

T25: retrofits and road-vehicle exchange rate, block-length scan, three-resource dock network. T26: floating docks, finite holding area S, K₀(S). Chapter 4 prototype: CP-SAT, LNS, T5 proof

RISK	MITIGATION
Safe release is conservative (2/3 of waits from refusals); reservation sets the lower edge for now	T27 adds the step-wise push and T5; then a space-time table (PBS, LNS) lifts the edge further
With detours, T1' (fixed loaded routes) is no longer a bound	Route-free T1" (T23) is the upper edge: route slack on Okpo's main scenario +3.7% (bound gap +5.3%), Yantai 0
CP-SAT solves only small instances	Small instances give ground truth; at scale, LNS with its gap to T1' and T1" reported
Field anchors differ in working day; Okpo's shifts are not public	Report 16 h and 24 h (at 24 h real fleets sit between free flow and reservation); the author sets the main caliber after T27
Okpo moves to map rev 1833 (floating docks, branch roads)	T27 step 0 reports the network and curve changes first; existing runs keep their registered maps

Target journals (paper 2, chapter 4): only journals in the school's 2026 directory

First choice (B)

- *IEEE Transactions on Automation Science and Engineering* (class B)
Deadlock avoidance and scheduling in automated handling; T5 alongside the algorithms
- *Transportmetrica A: Transport Science* (class B)
Online dispatching and network orchestration, with rigorous models and strong baselines

Reach (A)

- *IEEE Transactions on Industrial Informatics* (class A)
If rolling online orchestration is linked to a digital twin and closed with real or high-fidelity data
- *IEEE Transactions on Robotics* (class A)
If written as multi-agent planning for agents that change shape when loaded, with provable safety

Alternatives (B)

- *Engineering Applications of Artificial Intelligence* (class B)
Engineering mechanism, intelligent method, decision gains
- *Expert Systems with Applications* (class B)
When rules are learned or evolved by language models
- *IEEE Robotics and Automation Letters* (class B)
A short paper on multi-vehicle coordination and safe release

Supplementary (C)

- *Computers & Industrial Engineering* (class C)
- *Autonomous Agents and Multi-Agent Systems* (class C)
- *Journal of Manufacturing Systems* (class C)

B journals first; A journals are a reach, each with its condition. The school requires at least one paper in a B-or-above journal; paper 2 joins the other papers in the overall set.

References (1/2)

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Other: UIC (2013) *UIC Code 406: Capacity*, 2nd ed.; Google OR-Tools CP-SAT; KED Global (2024) on Hanwha Ocean transporters and signallers.

In short: from a reliable starting point, push the deadlock-free lower edge up

Grounds

The rule sets the interval's lower edge; the best deadlock-free rule is whole-route reservation at 36% of T1'; safe release is exact and deadlock-free but more conservative.

Approach

T27 first tries wider safe releases (step-wise push, T5); then assignment and passage order are searched together inside the shell; CP-SAT ground truth; rolling online.

Outputs

Chapter 4's rule comparison and deadlock-free lower edge, H5's cost of safety, a proof of T5, and a paper (first choice IEEE T-ASE or Transportmetrica A, both B).

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Where the network binds, schedule the road rather than the vehicles: orchestration decides how much of the bound is achieved.

Thank you.